

5781011

B.Tech. DEGREE EXAMINATION, MAY 2017.

First/Second Semester

ENGINEERING MECHANICS

(Common to All Branches)

(From 2013-14 Batch onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL the questions.

All questions carry equal marks.

1. State parallelogram law of forces.
2. Write the necessary conditions for equilibrium of a particle subjected to coplanar concurrent force system.
3. List the types of supports that are commonly used.
4. State the laws of Coulomb friction.

5. Define moment of inertia.
6. State Pappus-Guldinus theorem.
7. Determine the velocity with which a water jet must be projected vertically upwards to reach a height of 50 m.
8. Write the principle of conservation of momentum.
9. Find angular acceleration of a body, if its angular velocity is increased from 8 rad/s to 28 rad/s in 5 seconds.
10. State the principle of work and energy applicable for a rigid body.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the units Choosing ONE full questions from each Unit.

UNIT I

11. An automobile is pulled by means of two trucks as shown in figure 1. If the resultant of the two forces acting on the automobile is 20 kN being directed along the positive direction of X-axis, determine the angle ' θ ' of the cable attached to the truck at 'B' such that the force ' F_B ' in this cable is

minimum. What is the magnitude of force in each cable when this occurs?

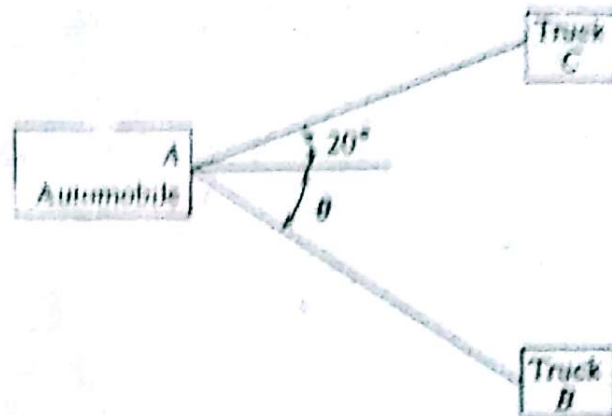


Fig 1
Or

12. Three-cables are used to support the 10 kg cylinder shown in the figure 2. Determine the force developed in each cable for equilibrium.

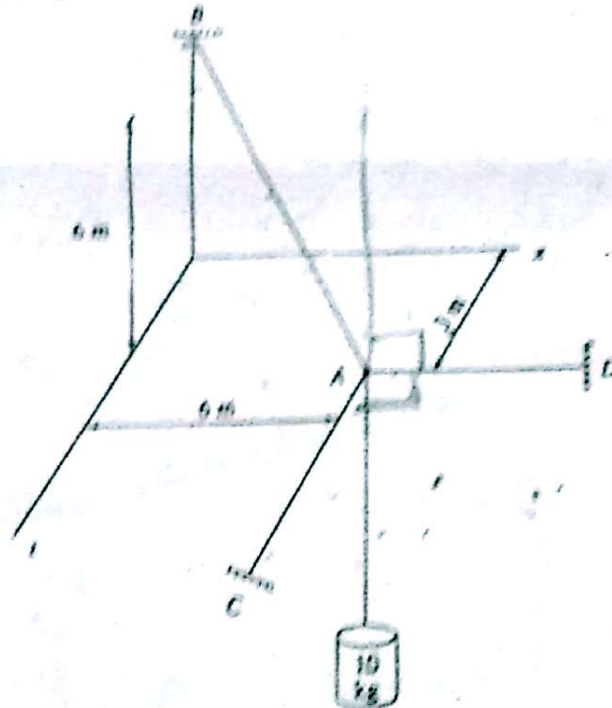


Fig 2

UNIT II

13. Determine the reactions at the supports A and B of the truss shown in figure 3.

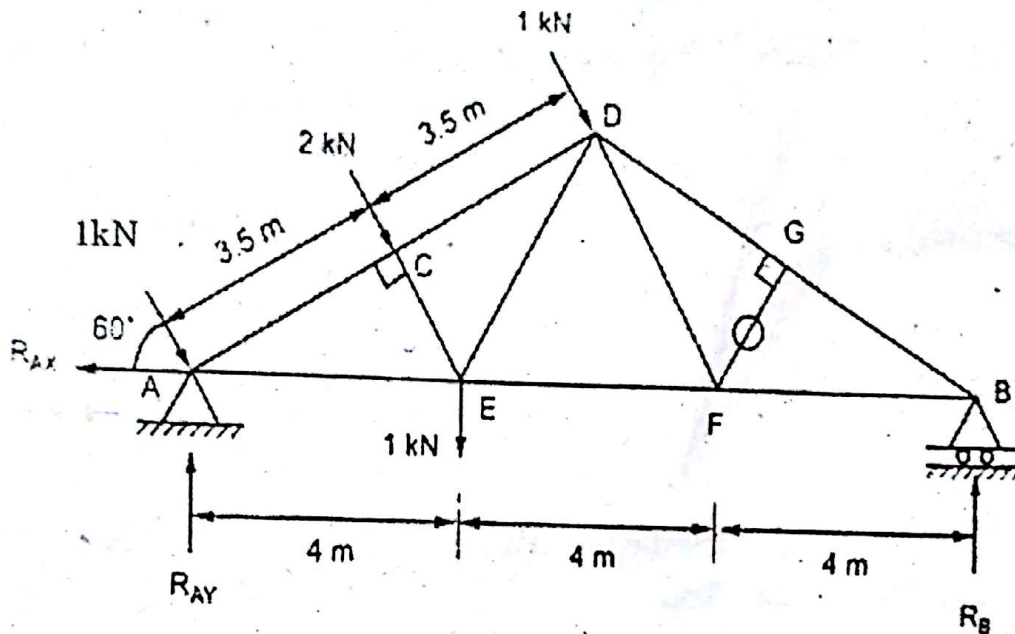


Fig. 3

Or

14. A body, resting on a rough horizontal plane required a pull of 180 N inclined at 30° to the plane just to move it. It was found that a push of 220 N inclined at 30° to the plane just moved the body. Determine the weight of the body and the coefficient of friction.

UNIT III

15. Find the area moment of inertia for the 'T' section shown in figure 4 about X-X axis and Y-Y axis passing through its centroid.

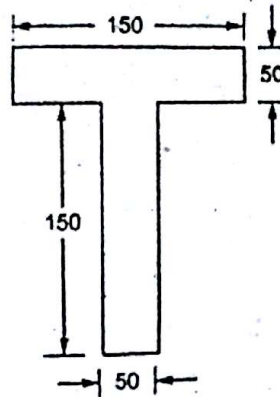


Fig 4

Or

16. Determine the centroid and area moment of inertia about its centroidal axis X-X & Y-Y for the given cut section figure 5

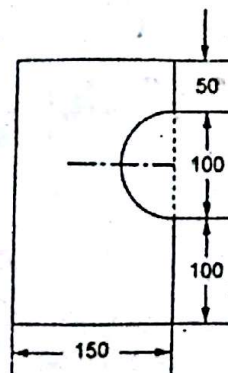


Fig 5

UNIT IV

17. A stone is thrown vertically upwards with velocity of 30 m/s from the top of a tower 50 m high. Calculate :

- (a) time required for the stone to reach the ground,
- (b) velocity of the stone in its downward travel at the point of throwing upwards and
- (c) maximum height to which the stone will rise in its flight.

Or

18. A vehicle of mass 600 kg and moving with a velocity of 10 m/s strikes another vehicle of mass 400 kg, moving at 9 m/s in the same direction. Both the vehicles get coupled together due to impact. Find the common velocity with which the two vehicles will move. Also find the loss of kinetic energy due to impact.

UNIT V

19. A flywheel weighing 50 kN having radius of gyration 1 m losses its speed from 400 rpm to 280 rpm in 2 minutes. Calculate:

- (a) retarding torque acting on it.
- (b) change in kinetic energy during the above period and
- (c) change in angular momentum during the same period.

Or

20. A 60 kg homogeneous cylinder 1000 mm in diameter accelerates from rest to a speed of 120 rev/min in just one revolution. Determine the torque which is needed to keep this speed. Also find the power required.
-

5781011

B.Tech. DEGREE EXAMINATION, NOVEMBER 2016.

First / Second Semester

ENGINEERING MECHANICS

(Common to All Branches)

(From 2013 – 14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define force-couple system.
2. Write the equations of equilibrium of a coplanar system of forces.
3. List the different types of loads acting on a structural member.
4. Define the coefficient of static friction.
5. When would the centroid and the center of mass coincide?
6. Find the radius of gyration of a rectangular area of MI about its base $9 \times 10^4 \text{ cm}^4$ and cross-sectional area 300 cm^2 .

7. Distinguish uniform motion and uniformly accelerated motion.
8. A body is moving with a velocity of 5 m/sec. After 3 sec, the velocity is 8 m/sec. Find the acceleration.
9. State the principle of work and energy.
10. Define absolute motion.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the Units, choosing ONE question from each Unit.

UNIT I

11. Find the magnitude, direction and the location of the resultant of the system of forces of the beam loaded as shown in Figure 1 (11)

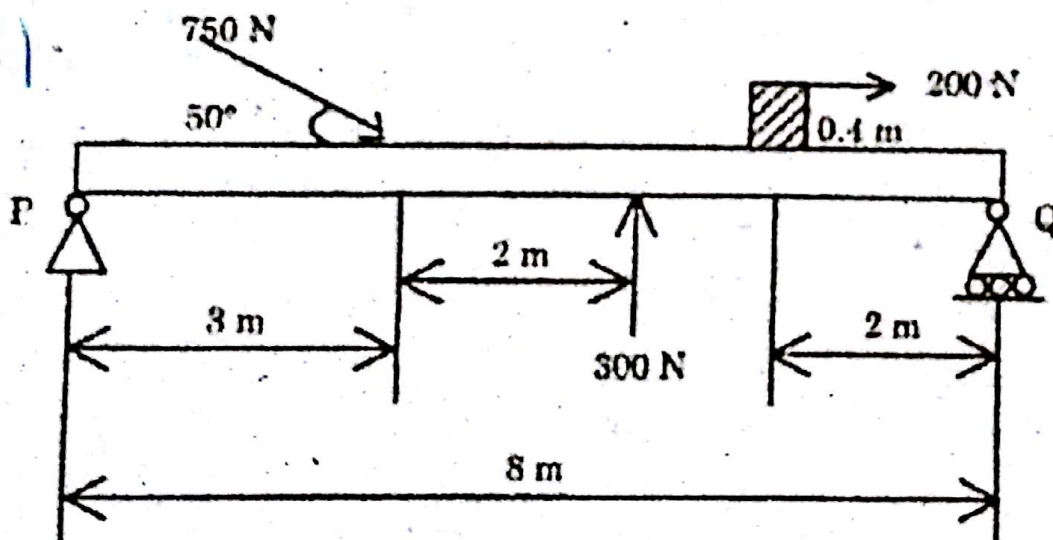


Figure 1

Or

12. Five forces 4 , $\sqrt{3}$, 5 , $\sqrt{3}$ and 3 kN respectively act at one of the angular points of a regular hexagon towards other five angular points. Find the magnitude and direction of the resultant and equilibrium forces. (11)

UNIT II

13. Determine the forces in all the members of the truss shown in Figure 2 and indicate the magnitude and nature of forces on the diagram of the truss. All inclined members are at 60° to the horizontal and length of each member is 2 m.

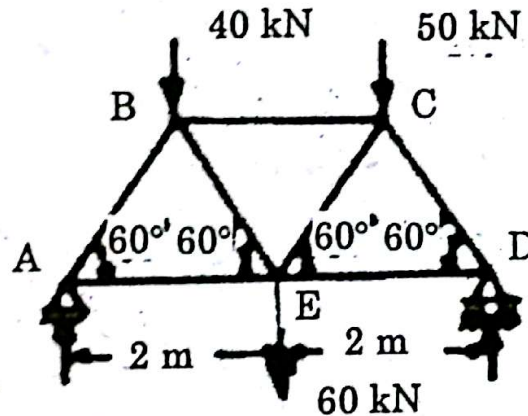
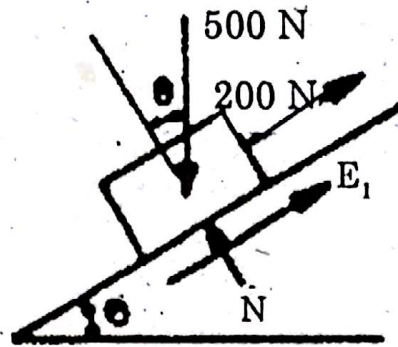


Figure 2

Or

14. A weight 500 N just starts moving down a rough inclined plane supported by a force of 200 N acting parallel to the plane and it is at the point of moving up the plane when pulled by a force of 300 N parallel to the plane. Find the inclination of the plane and the coefficient of friction between the inclined plane and the weight. (11)



UNIT III

15. Determine the moments of inertia and the radius of gyration of the shaded area, in Figure 3, with respect to the x and y axes. (11)

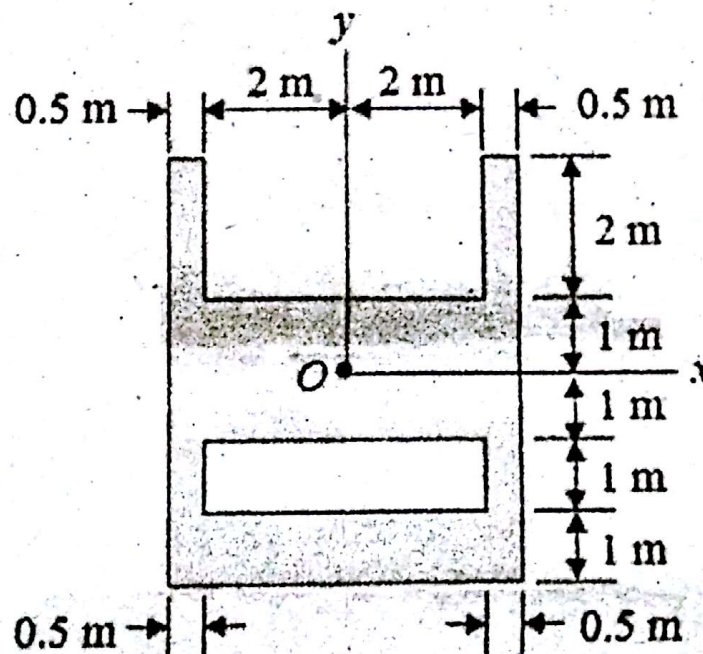


Figure 3

Or

16. Find the position of centroid of a frustum of a solid right circular cone base having an axial hole of 0.25 m diameter as shown in Figure-4. (11)

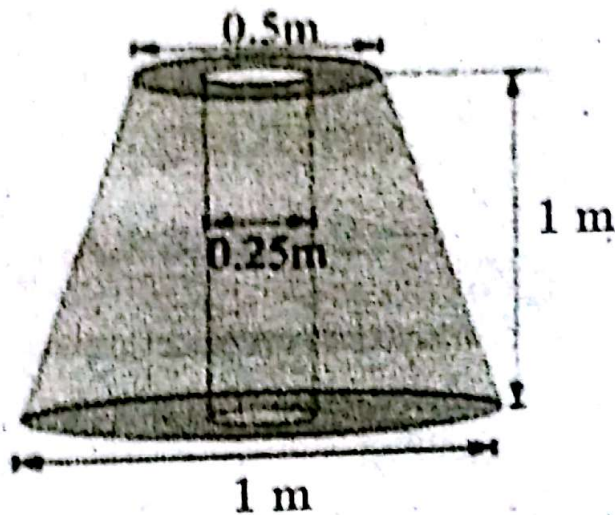


Figure 4

UNIT IV

17. A motorist travelling at a speed of 70 kmph suddenly applies brakes and halts after skidding 50 m. Determine,
- The time required to stop the car, and
 - The coefficient of friction between the tyres and the road. (11)

Or

18. A Golf ball is dropped from a height of 10 m on a fixed steel plate. The coefficient of restitution is 0.894. Find the height to which the ball rebounds on the first, second and third bounces. (11)

UNIT V

19. The angular acceleration of a flywheel is given by $\alpha = 12 - t$, where α is in rad/sec^2 and t is in sec. If the angular velocity of the flywheel is 60 rad/sec at the end of 4 sec. determine the angular velocity at the end of 6 sec. How many revolutions take place in these 6 sec? (11)

Or

20. Power supply was cut-off to a power driven wheel when it was rotating at a speed of 900 rpm . It was observed to come to rest after making 360 revolutions. Determine its angular retardation and time it took to come to rest after power supply was cut-off. (11)

5781011

B.Tech. DEGREE EXAMINATION, MAY 2016.

First/Second Semester

(Common to All Branches)

ENGINEERING MECHANICS

(From 2013–14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Write the concept of resolution of a plane force of Graphical methods.
2. Definitions of Concurrent Forces.
3. Write down the types of supports in two dimensions with sketches.
4. What is the type of load help of sketches?
5. Describe the second moment of area.
6. Describe the polar moment of Inertia.
7. Detail about the Kinematics and Kinetics.
8. Difference between the velocity and speed.
9. Short notes on Relative motion.
10. Explain the Linear Impulse and momentum.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL units, choosing ONE full questions from each Unit.

UNIT I

11. Find the magnitude and direction of the resultant of the concurrent forces of 80 N, 120 N, 150 N and 200 N making angles 30° , 70° , 120° , 15° and 155° respectively with X axis.

Or

12. Derived from the Lami's theorem.

UNIT II

13. Find the moment of the space force shown in fig.1 about the origin

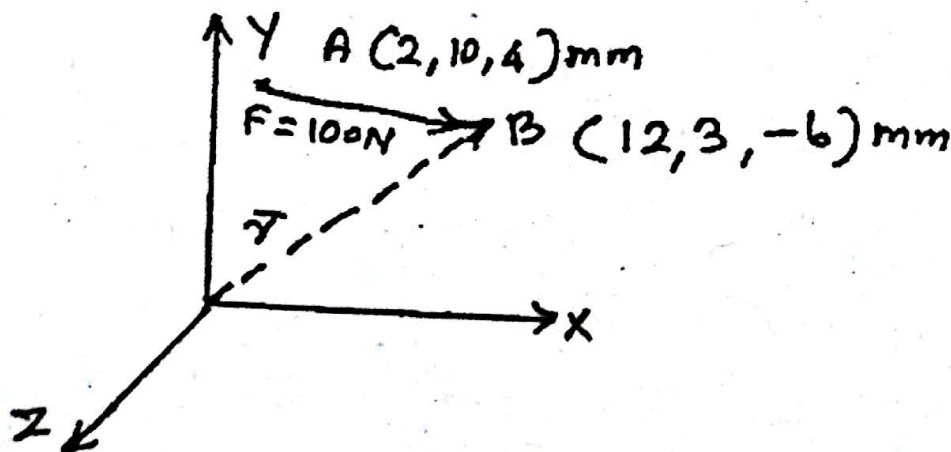


Fig-1

Or

14. Find the truss shown in fig. 2&3 Find the support reactions.

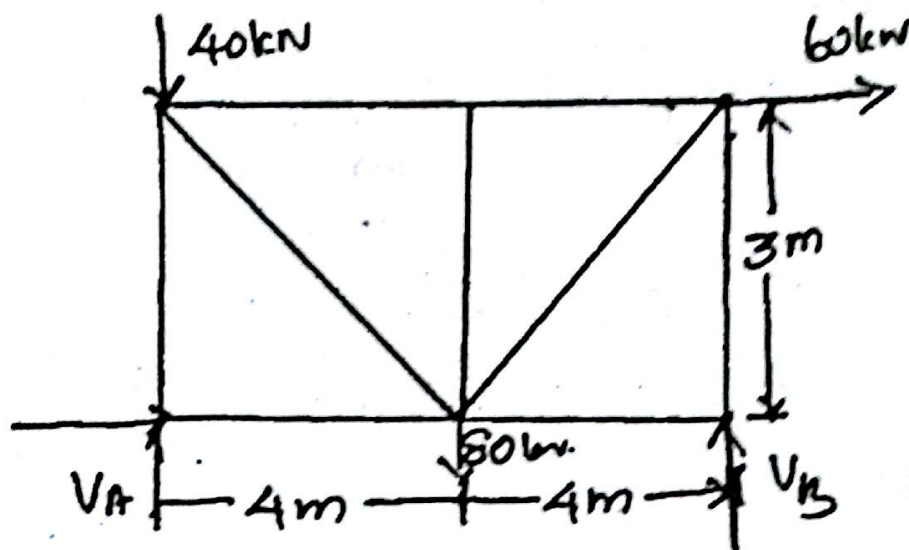


Fig. 2

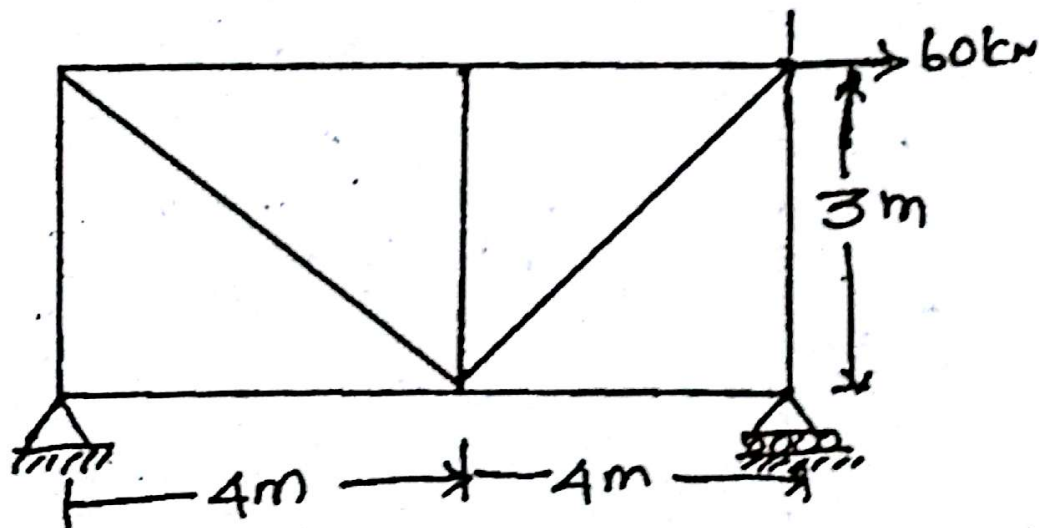


Fig. 3

UNIT III

15. Find the moment of inertia of the shaded area shown in fig. 4 about the centroidal XX and YY

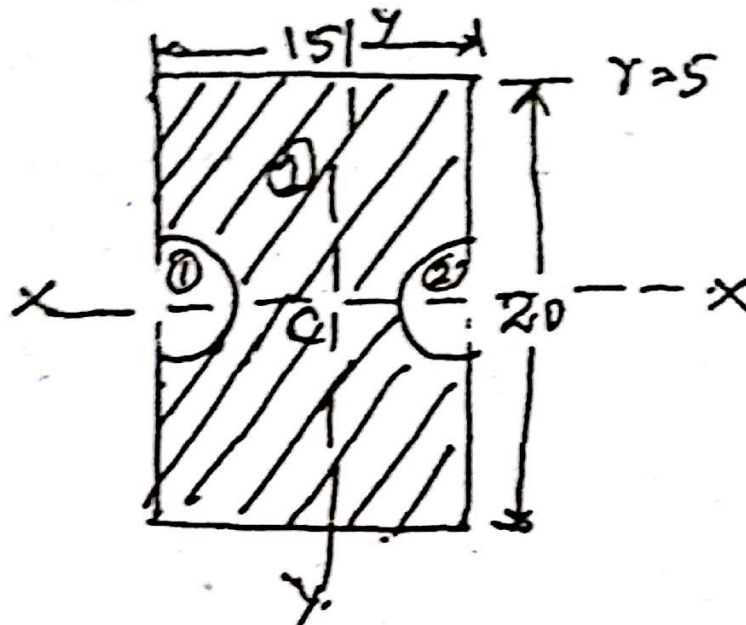


Fig. 4

Or

16. Find the moment of inertia of L angle ISA 100 × 75 × 6 about the centroidal XX and YY. (Fig.5)

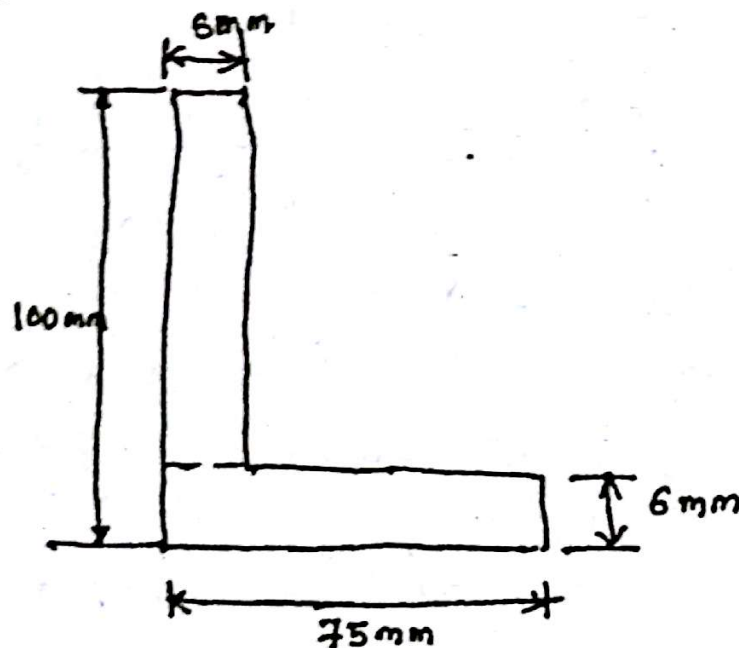


Fig. 5

UNIT IV

17. The equation of motion of a particle moving along a straight line is given by $x = 10t^3 - 30t^2 - 80$, what is the total distance travelled by the particle when the velocity changes from 30 m/s to 90 m/s. Also determine travelled during the 3rd seconds.

Or

18. A motor car takes 10 seconds to cover 20 m and another 15 seconds to cover another 40 m. Find the initial velocity and the constant acceleration. Also determine the velocity at the end of 15 seconds.

UNIT V

19. A stone is projected vertically upwards from the top of a building with a velocity of 35 m/s. Five seconds later another stone is just released from the same point. It is seen that both the stones reach the ground at the same time. Find the height of the building.

Or

20. A particle under constant deceleration is moving in a straight line and covers a distance of 20 meters in the first 2 seconds and 40 meters in the next 5 seconds. Calculate the distance it covers in the subsequent 3 seconds and the total distance travelled by the particle before it comes to rest.
-

5781011

B.Tech. DEGREE EXAMINATION, JANUARY 2016.

First/Second Semester

ENGINEERING MECHANICS

(Common to all branches)

(From 2009-10 onwards)

Time : Three hours

Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. State the principle of transmissibility.
2. Give the conditions for equilibrium of rigid bodies.
3. What do you mean by free body diagram?
4. Define angle of friction and angle of repose.
5. State parallel axis theorem.
6. Define radius of gyration.
7. State D' Alembert's principle.

8. Define coefficient of restitution.
9. What do you mean by general plane motion?
10. Define instantaneous centre of rotation.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, choosing ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Four coplanar forces are acting at a point as shown in figure 1. Three forces have magnitude of 20 N, 50 N and 20 N at angles of 45° , 200° and 270° respectively. Fourth force is unknown. Resultant force has magnitude of 50 N and acts along x-axis. Determine the unknown force and its direction from x-axis.

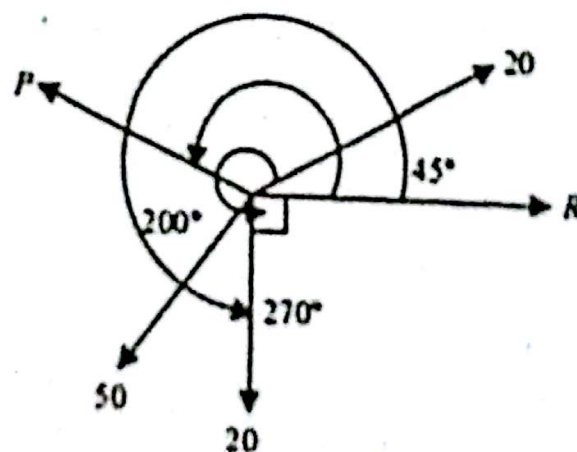


Figure 1

Or

12. A fine light string ABCDE whose extremity A is fixed, has weights W_1 and W_2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 40 N at the free end E as shown in figure 2. If in the position of equilibrium, BC is horizontal and AB and CD makes 150° and 120° with BC, find (a) Tension in the portion AB, BC and CD of the string and (b) Magnitude of W_1 and W_2 .

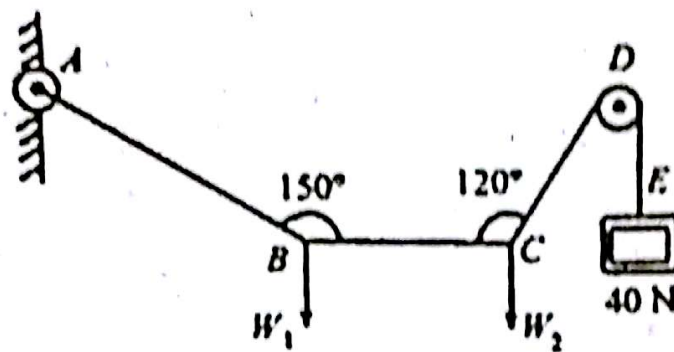


Figure 2

UNIT II

13. Find the reactions at supports A and B of the loaded beam shown in figure 3.

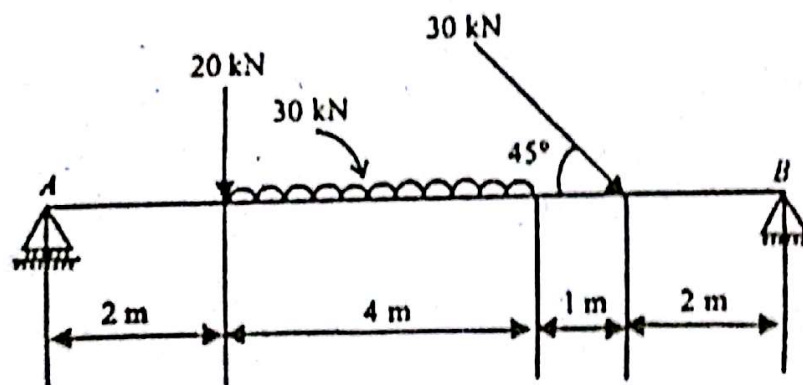


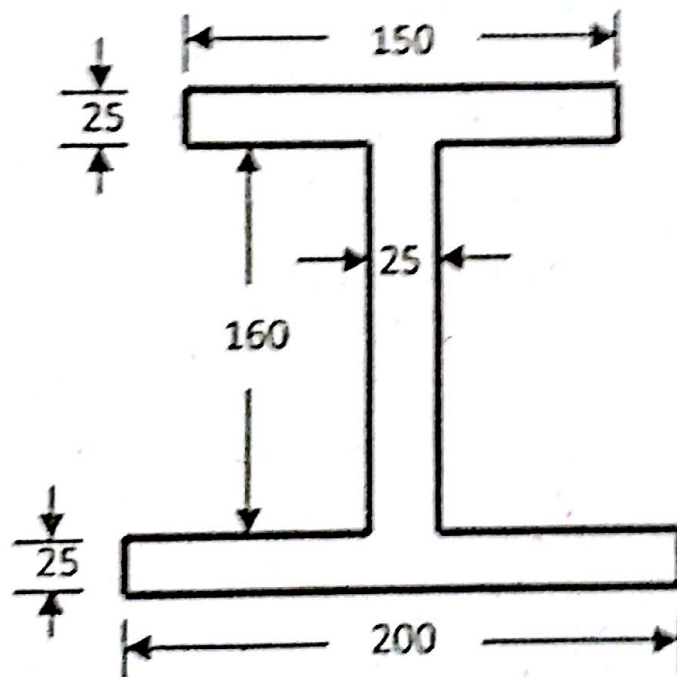
Figure 3

Or

14. Force of 200 N is required just to move a certain body up an inclined plane of angle 15° , the force being parallel to plane. If angle of inclination is made 20° the effort again required parallel to plane is found 250 N. Determine the weight of body and coefficient of friction.

UNIT III

15. Find center of gravity for the given 'I' section shown in figure 4.

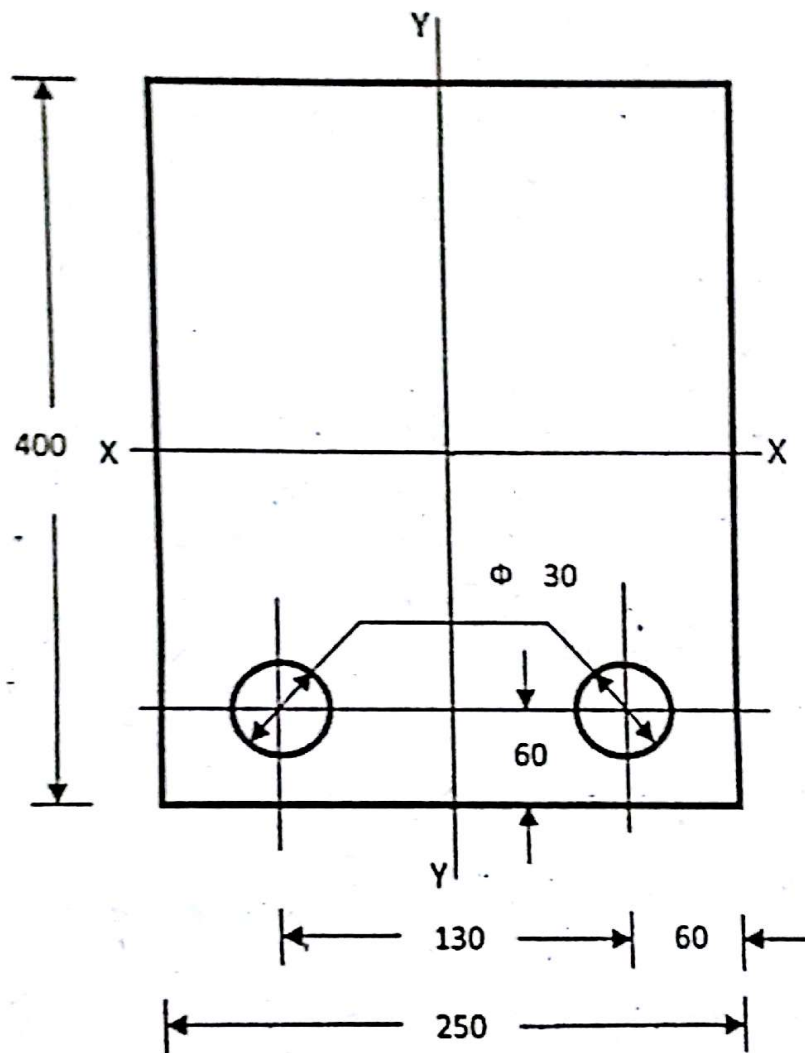


All dimensions are in mm

Figure 4

Or

16. Find area moment of inertia for the rectangle with two circular holes shown in figure 5 about the centroidal axes X-X and Y-Y.



All dimensions are in mm

Figure 5

UNIT IV

17. A bullet of mass 81 gm moving with a velocity of 300 m/s is fired into a log of wood and it penetrates to a depth of 10 cm. If the bullet moving with the same velocity were fired into a similar piece of wood 5 cm thick, with what velocity would it emerge? Find also the force of resistance assuming it to be uniform.

Or

18. A mass 12 kg travelling to the right with a speed of 7.5 m/s collides with another mass 24 kg, travelling to the left with a speed of 25 m/s. If the coefficient of restitution is 0.6, find the velocities of the particles after collision and loss in kinetic energy. What is the impulse acting on either particle during the impact?

UNIT V

19. A grinding wheel is attached to a shaft of an electric motor having a rated speed 3600 rpm. When the power is turned on the unit reaches its rated speed in 5 s and when the power is turned off the unit comes to rest in 70 s. Assuming uniformly accelerated motion, determine the number of revolutions made by the wheel (a) in reaching its rated speed, (b) in coming to rest.

Or

20. A block of weight 800 N is hung by using an inextensible cable that is wound around a cylindrical drum of radius 30 mm that is rigidly fixed to a flywheel. The combined moment of inertia of the flywheel and the drum about their centre of gravity is 12 kgm^2 . The moment developed due to bearing friction is 65 Nm. The velocity of the block at the instant considered is 1.5 m/s downward. Find the velocity of the block after it has moved 1 m.
-